

Supplementation of black rice pigment fraction improves antioxidant and anti-inflammatory status in patients with coronary heart disease.

Black rice and its pigment fraction have shown anti-atherogenic activities in several animal models, but whether their beneficial effects will recur in humans remains unknown. The aim of the present study is to investigate the influence of black rice pigment fraction (BRF) supplementation on selected cardiovascular risk factors in patients with coronary heart disease (CHD). Sixty patients with CHD aged 45-75 years were recruited from the Second Affiliated Hospital of Sun Yat-Sen University in Guangzhou, China and randomly divided into two groups. In the test group, the diet was supplemented with 10 grams of BRF derived from black rice for 6 months; While in the placebo group, the diet was supplemented with 10 grams of white rice pigment fraction (WRF) derived from white rice. At baseline, plasma antioxidant status and the levels of inflammatory biomarkers and other measured variables were similar between two groups. After 6 months' intervention, compared to WRF supplementation, BRF supplementation greatly enhanced plasma total antioxidant capacity (TAC) ($p=0.003$), significantly reduce plasma levels of soluble vascular cell adhesion molecule-1 (sVCAM-1) ($p=0.03$), soluble CD40 ligand (sCD40L) ($p=0.002$) and high sensitive C-reactive protein (hs-CRP) ($p=0.002$) in the test group. No significant changes were observed in plasma total superoxide dismutase (T-SOD) activity, lipids level and carotid artery intima-media thickness (IMT) between two groups. These results may suggest that BRF could exert cardioprotective effects on patients with CHD by improving plasma antioxidant status and inhibiting inflammatory factors.